

How Much Adaptation Is Enough?

Drift Geometry and Minimal Calibration in Brain-to-Text BCI

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Abstract

Brain-to-text BCI decoders must remain usable as neural signals change across recording sessions. We audit how much input adaptation is needed when the official T15 brain-to-text GRU is evaluated across sessions. Native day-specific input layers reproduce low validation phoneme error (9.06% PER), while forcing sessions through non-native input layers increases PER to 22.67–34.43%. Unsupervised moment corrections fail to recover this loss. A supervised input-layer adapter trained from 20 labeled calibration trials recovers 22.7% of the native-day gap, improving over shorter adapter training but still leaving substantial degradation. The strongest practical result is source-layer selection: using a short unlabeled beginning-of-day window to choose a past input layer recovers much of the fixed-source loss, but the immediately previous session is an even stronger simple baseline. Secondary geometry-only analyses of T12 diagnostic blocks and inner-speech behavior blocks show similar state structure beyond the main T15 decoder experiment. These results suggest that recency should be the default adaptation prior, while drift geometry is a useful signal for deciding when that default may need to be overridden.

1 Introduction

High-performance speech BCIs commonly include session-specific adaptation machinery because intracortical neural features drift across days. In this work, “drift” means that the neural feature distribution recorded on a new day no longer matches the distribution seen by the decoder during earlier sessions. A practical BCI therefore needs a way to decide whether yesterday’s input mapping is still usable, whether an older mapping is a better match, or whether a short calibration step is needed. This raises a practical question: before retraining a decoder or collecting a long supervised calibration set, how much adaptation is enough?

We examine this question in the official T15 brain-to-text GRU baseline from a rapidly calibrating speech neuroprosthesis study [Card et al., 2024]. The model contains learned day-specific input layers, followed by a GRU and output head. Intuitively, each input layer acts like a small session-specific adapter that maps that day’s neural features into the representation expected by the shared GRU decoder. This structure creates a useful stress test: if a target session is decoded through the wrong input layer, performance measures the decoder-facing cost of cross-session input mismatch. We then ask whether simple feature corrections, small learned adapters, or geometry-guided source selection can recover the lost performance.

The contribution is not a new speech decoder. Instead, we provide a decoder-facing adaptation audit:

- Non-native input layers cause a large decoder-facing degradation in T15, increasing PER from 9.06% native-day to 22.67–34.43% under fixed non-native input layers.
- Mean/variance correction is not enough: target z-scoring and moment matching remain near 22.7% PER, while input-layer calibration from $K = 20$ trials improves the fixed-middle stress setting to 18.59% PER but remains incomplete.
- Geometry-guided source selection recovers much of the fixed-source degradation: at $K = 20$, K-shot geometry reaches 12.74% PER, while previous-session recency is even stronger at 12.21%.
- T12 diagnostic blocks and inner-speech behavior blocks provide geometry-only support, including positive T12 days–distance correlations and 96.15% nearest-behavior accuracy in inner-speech blocks.

We do not claim that neural drift geometry is new. Rather, we show that in a deployed-style brain-to-text decoder with learned day-specific input layers, short-window geometry is actionable for choosing among existing input-layer states and for framing recency as an explicit adaptation prior.

80	2 Related Work	
81	2.1 Brain-to-Text BCIs	
82	Intracortical communication BCIs have advanced	
83	rapidly from handwriting-based brain-to-text decod-	
84	ing [Willett et al., 2021; Fan et al., 2023] to speech neu-	
85	roprostheses that decode attempted speech into text or	
86	spoken output [Willett et al., 2023; Card et al., 2024].	
87	These systems commonly rely on recurrent neural de-	
88	coders and session-specific adaptation or recalibration	
89	to maintain performance as recordings change over time.	
90	Recent work also explores transferring pretrained speech	
91	representations to brain decoding, for example by adapt-	
92	ing Wav2Vec2-style models to neural features [Fiedler et	
93	al., 2025]. That direction asks how stronger speech back-	
94	bones or pretrained acoustic representations can improve	
95	neural decoding. Our work is complementary: given an	
96	already trained brain-to-text decoder with day-specific	
97	input layers, we ask how to keep the decoder usable	
98	across sessions using lightweight input-side adaptation	
99	rather than retraining the backbone.	
100	2.2 Cross-Session Stability and Neural Geometry	
101	Cross-session stability has been studied through the lens	
102	of neural geometry. Manifold-alignment methods can	
103	stabilize fixed decoders by aligning low-dimensional neu-	
104	ral activity across recording instabilities [Degenhart et	
105	al., 2020], and recent dynamics-aware approaches such	
106	as NoMAD use temporal structure to perform unsuper-	
107	vised alignment over weeks to months [Karpowicz et al.,	
108	2025]. Our analysis uses this geometry perspective in	
109	a narrower decoder-facing setting: we measure whether	
110	short-window geometry can guide selection among exist-	
111	ing day-specific input-layer states.	
112	2.3 Session-Invariant and Continual Adaptation	
113	In speech BCI specifically, a recent preprint proposes	
114	adversarial session-invariant learning to reduce session-	
115	specific information while preserving phoneme decoding	
116	structure [Zhang et al., 2026]. Continual recalibration	
117	with pseudo-labels provides a complementary route, us-	
118	ing language-model corrections to update communica-	
119	tion decoders online [Fan et al., 2023]. Our work does	
120	not propose a new invariant decoder or full recalibra-	
121	tion method. Instead, it audits low-cost choices available	
122	before retraining: moment correction, lightweight input	
123	calibration, previous-session reuse, and geometry-guided	
124	source-layer retrieval.	
125	3 Data and Decoder	
126	T15. The main experiments use T15 copy-task neu-	
127	ral features from the official brain-to-text dataset [Card	
128	et al., 2024]. We evaluate 41 validation sessions with	
129	ground-truth phoneme labels. Neural inputs are passed	
130	through the official GRU checkpoint and decoded greed-	
131	ily at the phoneme level. During evaluation, repeated	
132	symbols and blank tokens are removed using the stan-	
133	dard CTC collapse rule before computing PER. This col-	
	lapse step is only part of decoding/evaluation; learned	134
	calibration is trained separately using CTC loss.	135
	Official baseline. The decoder contains learned day-	136
	-specific input layers, a GRU backbone, and an output	137
	layer over phoneme classes. In the native-day condition,	138
	each session uses its own input layer. In cross-day stress	139
	tests, target sessions use a fixed or selected non-native	140
	source input layer while the rest of the decoder remains	141
	unchanged.	142
	T12 diagnostic blocks. As a secondary check, we ana-	143
	lyze diagnostic blocks from the T12 high-performance	144
	speech neuroprosthesis dataset [Willett et al., 2023].	145
	This is a geometry-only analysis of spike-power and	146
	threshold-crossing feature variants; we do not reproduce	147
	the T12 decoder or report T12 PER.	148
	4 Methods	149
	4.1 Decoder-Facing Metrics	150
	For each T15 validation trial, we compute greedy	151
	phoneme predictions from the CTC output stream and	152
	then apply CTC collapse before computing phoneme er-	153
	ror rate (PER). We report phoneme-weighted PER un-	154
	less otherwise stated. Quality diagnostics such as blank	155
	rate, confidence, and entropy are saved, but the core re-	156
	sults use PER.	157
	For a set of trials $\mathcal{D}$, the reported PER is	158
	$\text{PER}(\mathcal{D}) = \frac{\sum_{i \in \mathcal{D}} \text{edit}(y_i, \hat{y}_i)}{\sum_{i \in \mathcal{D}} y_i }.$	
	When comparing an adaptation method to a fixed-source	159
	stress test, we report the fraction of the fixed-source gap	160
	recovered:	161
	$\text{Recovery} = \frac{\text{PER}_{\text{fixed}} - \text{PER}_{\text{method}}}{\text{PER}_{\text{fixed}} - \text{PER}_{\text{native}}}.$	
	This quantity is useful because the learned-adaptor ex-	162
	periments evaluate on remaining validation trials after	163
	holding out the first K calibration trials, whereas the	164
	native-day and fixed-source probes can also be reported	165
	on the full validation split.	166
	4.2 Cross-Day Stress Test	167
	We compare native-day decoding against three fixed	168
	non-native source layers: early (t15.2023.08.13), mid-	169
	dle (t15.2023.11.26), and late (t15.2025.04.13). This	170
	tests whether the decoder degradation is tied to one bad	171
	source layer or is a general consequence of input-layer	172
	mismatch.	173
	4.3 Adaptation Ladder	174
	Using the middle source layer, we test no correction, tar-	175
	get z-scoring, source moment matching, diagonal affine	176
	calibration, and full input-layer calibration. Moment	177
	methods use full-session unsupervised feature statistics.	178
	Learned adapters use the first K labeled validation trials	179
	and are evaluated only on the remaining trials. We use	180
	K to simulate a beginning-of-day calibration window:	181

the first K trials are used for adaptation, and the rest of the session is held out for evaluation.

The two unsupervised moment corrections are deliberately simple. Target z-scoring maps x to $(x - \mu_t)/(\sigma_t + \epsilon)$, while source moment matching maps target features into the source marginal statistics:

$$x' = \frac{x - \mu_t}{\sigma_t + \epsilon}(\sigma_s + \epsilon) + \mu_s.$$

These baselines test whether the day-specific input layers can be approximated by channel-wise distribution alignment. Diagonal affine and input-layer calibration are supervised small-calibration probes: the GRU and output head remain frozen, and only a small input transform is learned from the first K validation trials using CTC loss.

4.4 Drift Geometry and Source Selection

For each source-target pair, we compute temporal distance, mean and scale shifts, relative covariance shift, CORAL-style distance, and PCA subspace distances. We then select a source input layer by nearest covariance geometry. To approximate deployment, a K-shot variant estimates target geometry from only the first $K \in \{5, 10, 20\}$ unlabeled validation trials and evaluates on the remaining trials. We focus the main claims on $K \leq 20$ because this represents the minimal-calibration regime in the available validation split.

Because temporal recency is an obvious baseline, we also evaluate the immediately previous input layer. Finally, we test a simple recency-aware override: use the previous input layer by default and switch to the geometry-selected older source only if its covariance distance is less than α times the distance to the previous source.

The K-shot source-selection rule is unsupervised with respect to labels. Let $S_t^{(K)}$ denote statistics estimated from the first K target trials, and let S_s denote source-session statistics estimated from past data. The geometry-selected source is

$$s^*(t) = \arg \min_{s \in S_{\text{past}}(t)} d_{\text{cov}}(S_t^{(K)}, S_s),$$

where d_{cov} is relative Frobenius covariance shift. We compare this to a previous-session policy, $s_{\text{prev}}(t)$, because recency is the strongest prior available in a longitudinal BCI.

4.5 Input-State Library Policies

The source-selection experiments can be interpreted as retrieving an input state from a memory of previous sessions. We therefore compare four policies: using only the previous session, selecting geometrically from the last 3 sessions, selecting from the last 5 sessions, and selecting from all past sessions. This ablation asks whether a practical system needs to retain all historical input layers, or whether a bounded recent library is enough under the current geometry metric.

Table 1: T15 decoder-facing stress test and adaptation ladder. Fixed-source and moment-correction rows use the middle source t15.2023.11.26. Learned adapter rows use $K = 20$ calibration trials and are evaluated on remaining trials. Lower PER is better.

Condition	PER	Recov.	Note
Native-day	9.06%	100%	correct layer
Fixed early	28.81%	—	40/41 harmed
Fixed middle	22.67%	0%	stress source
Fixed late	34.43%	—	40/41 harmed
Target z-score	22.75%	-0.6%	fails
Moment match	22.73%	-0.4%	fails
Diag. affine, $K = 20$	20.21%	10.1%	adapter
Input layer, $K = 20$	18.59%	22.7%	20 epochs

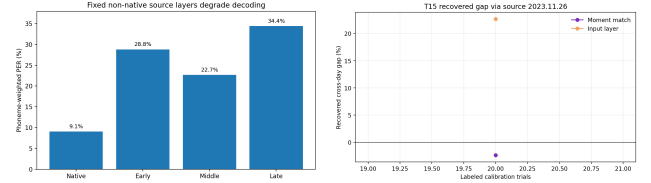


Figure 1: Left: fixed non-native source layers strongly degrade T15 phoneme decoding relative to native-day decoding. Right: small learned calibration recovers only a modest fraction of the fixed-source gap.

5 Experiments

We organize the empirical study into six experiments, each paired with a table or figure. The T15 experiments are the main decoder-facing results; T12 and inner-speech analyses are supporting geometry-only checks. The experiments follow an adaptation ladder: fixed source $\rightarrow$ moment correction $\rightarrow$ learned calibration $\rightarrow$ source selection $\rightarrow$ recency/library policies $\rightarrow$ geometry-only supporting checks.

5.1 Cross-Day Stress Test and Adaptation Ladder

Native-day decoding reaches 9.06% phoneme-weighted PER, confirming that the local evaluation reproduces the expected behavior of the official checkpoint. When all sessions are forced through non-native source input layers, PER rises to 22.67–34.43%, harming 40/41 sessions for each source choice.

The natural first hypothesis is that cross-session mismatch is mostly a mean/variance shift. The data do not support this. Target z-scoring and moment matching to the source distribution remain essentially tied with no correction, and are slightly worse in aggregate. This suggests that the learned day-specific input layers capture transformations richer than channel-wise mean and variance.

Supervised diagonal affine and input-layer calibration improve PER beyond moment matching. With $K = 20$, diagonal affine reduces weighted PER from 21.51% to 20.21%. A longer CUDA input-layer calibration run improves the learned input layer to 18.59%, recovering 22.7% of the native-day gap. A follow-up initial-

Table 2: Beginning-of-day source selection on remaining validation trials. Previous-session recency is the strongest simple baseline, while K-shot geometry selection remains close and often selects non-previous sources. Lower PER is better.

Method	$K = 5$	$K = 10$	$K = 20$
Native-day	8.95%	8.88%	8.90%
Fixed middle	22.51%	22.40%	22.30%
Previous source	13.09%	12.84%	12.21%
K-shot geometry source	14.00%	13.19%	12.74%
Best simple override	13.09%	12.90%	12.18%

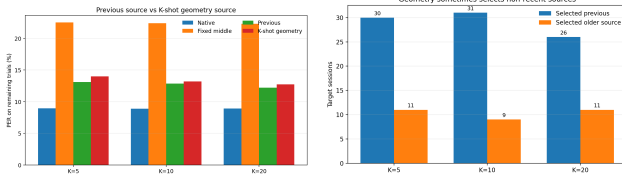


Figure 2: Left: previous-session recency is a strong source-selection baseline and slightly outperforms unconditional K-shot geometry selection. Right: geometry still reveals non-recent source matches, suggesting sessions are not always closest to the immediately previous day.

ization check suggests that calibration should be treated as residual adaptation rather than reset: on the same $K = 20$ subset, initializing from the previous input state improves 11.58% to 11.03%, whereas fixed-middle initialization remains much weaker at 17.93%. Thus learned input adaptation is a useful rung of the adaptation ladder, but even this stronger small-calibration setting does not replace a well-matched prior input state.

5.2 Beginning-of-Day Source Selection

Geometry-guided source selection gives a much stronger recovery than directly adapting a fixed middle source. In an offline all-candidate setting, choosing the nearest non-native input layer by covariance geometry reaches 11.38% PER. In a stricter past-only setting, it reaches 13.58% PER on 40 sessions, compared with 22.76% for the fixed middle source on the same sessions.

The beginning-of-day version is more realistic. Here, K denotes the first K beginning-of-day trials used for calibration or geometry estimation; evaluation is performed only on the remaining trials. With this protocol, past-only source selection reaches 14.00%, 13.19%, and 12.74% PER at $K = 5, 10, 20$, recovering 62.8%, 68.1%, and 71.3% of the fixed-source loss. However, the immediately previous input layer is even stronger: 13.09%, 12.84%, and 12.21% PER on the same subsets. This prior has a time horizon: previous-source PER is 9.97% when the last available input state is 0–3 days old, 13.75% at 4–14 days, and 24.54% beyond two weeks. Thus previous is a strong default only when it is truly recent.

Table 3: Evidence for non-recent source matches. These examples do not prove exact recurrence of neural states, but motivate maintaining a library of past input-layer states rather than only the most recent layer.

Data	Setting	Evidence	Interpretation
T15	K-shot	minority wins	override signal
T15	fingerprint	lower subspace angle	drift type
T12	diagnostic	08-13 → 06-21	53-day match
T12	diagnostic	08-25 → 08-13	older over previous

5.3 Reusable Input-State Libraries

The previous-session input layer is the strongest simple baseline, but it is not always the geometrically nearest source. In T15, K-shot geometry selection chooses the previous session in most targets, but selects an older non-previous input layer in 22–30% of cases. Depending on whether improvement is summarized by trial-mean PER or phoneme-weighted session PER, these older sources outperform the previous-session source in a small minority of non-previous selections. This suggests that some target sessions may resemble older input-layer states rather than only drifting monotonically away from the previous day.

One possible explanation is that a day-specific input layer may represent a neural recording state rather than a purely chronological day. Changes in recording quality, electrode yield, thresholding, neural strategy, or user state could move a new session closer to an older feature geometry than to the immediately preceding session. In this view, source selection is closer to memory retrieval over past input states than to choosing the most recent date.

We observe the same qualitative pattern in the T12 diagnostic-block geometry check. Using spike-power features, geometry selects an older non-previous session in 7/15 evaluable targets. Some selections jump substantially backward in time, for example 2022-08-13 selecting 2022-06-21, a 53-day lag, and 2022-08-25 selecting 2022-08-13 rather than the immediately preceding session.

The practical implication is that a longitudinal BCI should not keep only the latest input layer. A more robust system could maintain a small library of past input-layer states, use the previous session as the default, and invoke a recency-aware geometry gate only when the beginning-of-day neural geometry strongly suggests that an older state is a better match.

We therefore asked how large this past-state library needs to be. For each target session, we restricted K-shot geometry selection to the previous session, the last 3 sessions, the last 5 sessions, or all past sessions. The immediately previous session remains the best aggregate policy, but a small recent library is close to the all-past library. At $K = 20$, last-5 and all-past selection both reach 12.74% PER, while last-3 gives 12.76%. Expanding the nearest-source search to all past layers does not improve aggregate PER over recency under this simple distance rule; rather, it increases the need for a confidence gate before using older states.

Table 4: K-shot input-state library-size ablation. PER is measured on remaining validation trials after using the first K trials to estimate target geometry. A bounded recent library nearly matches all-past selection, while previous-only recency remains the strongest aggregate baseline.

K	Previous	Last 3	Last 5	All past
5	13.09%	13.47%	13.55%	14.00%
10	12.84%	13.19%	13.19%	13.19%
20	12.21%	12.76%	12.74%	12.74%

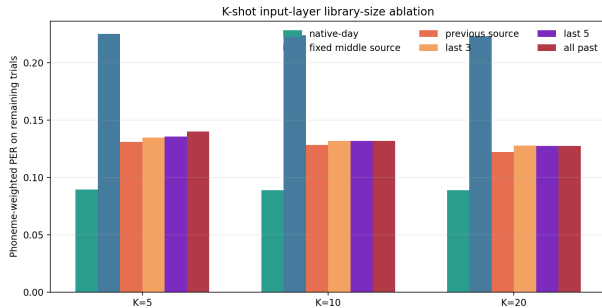


Figure 3: Library-size ablation. A small recent library approaches all-past geometry selection, but previous-session recency remains the strongest aggregate policy under the current covariance-nearest rule.

5.4 Gating and Drift Fingerprints

This experiment asks why geometry is useful but hard to deploy: can it safely override recency, and when do older states actually help? Since geometry sometimes chooses older sources, we tested whether a simple distance-ratio gate can improve over recency: keep the previous source unless the geometry-selected source is at least an α fraction closer in covariance distance. This rule does not yet produce a meaningful improvement. At $\alpha = 0.9$, it ties previous at $K = 5$, is slightly worse at $K = 10$ (12.90% vs 12.84%), and improves only marginally at $K = 20$ (12.18% vs 12.21%). At $\alpha = 1.0$, it approaches unconditional geometry selection and becomes worse. Geometry is therefore a useful candidate signal, but a deployable gate needs richer confidence features.

To test whether this failure is merely due to a poor hand-chosen threshold, we also train a tiny recency-aware gate. For each K , we fit a leave-one-session-out decision stump over geometry features such as distance ratio, absolute margin, source lag, and previous/source distances. This learned gate improves over previous-session recency only at $K = 20$, and only slightly (12.21% to 12.15% PER; Table 5). At $K = 5$ and $K = 10$, it overuses older sources and performs worse than previous. A one-dimensional learned gate is therefore still not enough; future gates need richer confidence signals.

The non-previous selections explain why the problem is difficult. Older sources beat the previous source in only a minority of non-previous geometry selections (3/11, 4/9, and 3/11 for $K = 5, 10, 20$ using phoneme-

Table 5: Leave-one-session-out learned gate. The gate defaults to the previous source and learns a one-feature decision stump for when to override with the geometry-selected older source. It gives only a tiny gain at $K = 20$. Lower PER is better.

K	Previous	Geometry	Gate	Overrides
5	13.09%	14.00%	13.37%	6
10	12.84%	13.19%	13.18%	5
20	12.21%	12.74%	12.15%	4

Table 6: Preliminary drift fingerprints for non-previous source choices. Negative principal-angle change means the older source is closer to the target subspace than the previous source.

K	Case	n	Δ principal angle
5	older wins	3	-1.37°
5	older loses	8	-0.06°
10	older wins	4	-1.93°
10	older loses	5	$+1.02^\circ$
20	older wins	3	-2.91°
20	older loses	8	$+0.55^\circ$

weighted session PER), and the Spearman correlation between geometry/previous distance ratio and older-source gain is weak ($\rho = 0.12, -0.02, 0.28$). Thus covariance geometry is useful for candidate retrieval, but not yet reliable as a standalone confidence measure.

To ask why older sources sometimes beat the previous session, we compare drift fingerprints for target–previous pairs and target–older-source pairs. This analysis is preliminary because older-source wins are rare, but it reveals a consistent pattern: wins are better explained by subspace geometry than by mean/scale shifts alone. When older sources win, they reduce the mean principal angle relative to the previous source by 1.37° , 1.93° , and 2.91° for $K = 5, 10, 20$. When older sources lose, this subspace improvement is absent or reverses (Table 6). This suggests that non-recent input states are useful when they better match the target neural subspace, supporting a drift-typing view of the adaptation ladder.

5.5 T12 Diagnostic Blocks Support the Geometry/Recency Framing

T12 diagnostic blocks provide a secondary geometry-only check. Using spike-power features, covariance distance increases with temporal separation across 120 past-source pairs (Spearman $\rho = 0.44$, $p = 5.4 \times 10^{-7}$). Geometry selects the immediately previous session in 8/15 target sessions and an older source in 7/15. Some selections jump far back in time, such as a 53-day lag. This pattern is not limited to spike power: threshold-crossing variants tx1–tx4 also show positive days-distance correlations ($\rho = 0.34$ – 0.47 , all $p < 1.3 \times 10^{-4}$) and older-source selections in 6–9 of 15 targets. This supports the idea that recency is important

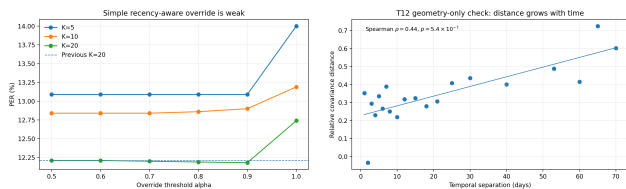


Figure 4: Left: a naive recency-aware geometry override does not substantially beat the previous-session baseline. Right: T12 diagnostic-block covariance distance increases with temporal separation, supporting the broader geometry framing outside T15.

Table 7: T12 diagnostic-block feature robustness. Across spike-power and threshold-crossing variants, covariance distance increases with temporal separation and geometry sometimes selects older non-previous sessions. This is a geometry-only supporting check.

Feature	$\rho(\text{days, cov.})$	Previous	Older
spikePow	0.44	8/15	7/15
tx1	0.35	7/15	8/15
tx2	0.34	6/15	9/15
tx3	0.34	9/15	6/15
tx4	0.47	9/15	6/15

but not sufficient to describe longitudinal neural geometry. Because we do not run the T12 decoder, this result is only supporting evidence, not decoder replication.

5.6 Inner-Speech Blocks Suggest Behavior States

As a geometry-only extension, we also analyzed interleaved verbal-behavior blocks from the T12/T15/T16/T17 inner-speech dataset [Kunz et al., 2025]. We did not train a decoder, evaluate PER, or attempt privacy-sensitive inner-speech decoding. Instead, within each participant, we grouped neural activity by speech behavior and split each behavior into two halves. We then asked whether covariance geometry retrieves the matching behavior. For binned threshold crossings during the go epoch, nearest-behavior accuracy was 25/26 (96.15%), compared with a 20% chance level, and between-behavior distances were 2.60 times within-behavior distances. Spike-power features and delay-epoch threshold crossings also remained well above chance (Table 8). These preliminary results suggest that the input-state view may extend beyond recording days: intended speech behavior is another axis of neural geometry.

6 Discussion

The main lesson is that cross-session adaptation should not be treated as a single fixed correction. The native T15 input layers are highly effective, and using the wrong input layer causes large decoder-facing degradation. But simple moment alignment does not emulate those layers, and learned adapters recover only part of

Table 8: Preliminary inner-speech behavior-geometry feasibility. This analysis is a supporting geometry-only check and does not evaluate decoding.

Feat./epoch	Acc.	Chance	B/W ratio
binnedTX / go	96.15%	20%	2.60
spikePow / go	84.62%	20%	2.87
binnedTX / delay	88.46%	20%	1.59

Table 9: Deployment interpretation of the adaptation ladder. The goal is not to apply every adaptation step every day, but to choose the lightest action that is supported by the beginning-of-day evidence.

Situation	Suggested action	Evidence in this audit
Recent state	Reuse previous	strongest baseline
Fixed-source mismatch	Avoid fixed source	22.67–34.43% PER
Mean/scale shift only	Moment correction insufficient	~22.7% PER
Labeled K trials	Calibrate input	partial recovery
Non-recent match	Retrieve older state	lower subspace mismatch
Behavior shift	Index behavior	inner-speech geometry

the lost performance even when input-layer calibration is trained longer.

A practical interpretation is a beginning-of-day triage procedure. The system should first try the previous input state, because it is the strongest simple default when it is truly recent. If short-window geometry suggests that the new session resembles an older stored state, the system can retrieve that state. If no stored state appears reliable, lightweight residual calibration can update the input mapping while leaving the GRU backbone and output head frozen.

Table 9 summarizes the practical interpretation: the system should not perform the same update every day. Instead, it should start from the cheapest reliable action and escalate only when the beginning-of-day evidence suggests that the current input state is no longer reliable.

The most useful low-cost intervention is not a new adapter, but a better choice of existing input layer. A short unlabeled beginning-of-day window is enough to choose a past layer that recovers much of the fixed-source loss. Yet the strongest simple policy is often even simpler: use the previous session. This changes the adaptation ladder. Recency should be the default prior; geometry should be used as an additional signal for detecting when the previous session is not the right source. This also suggests a practical memory mechanism: instead of overwriting old session adapters, the system should retain a library of past input-layer states and learn when the current session has returned to one of them.

The failed distance-ratio override is also informative. It shows that geometry is not automatically a better policy than recency. The drift-fingerprint analysis suggests that the next gate should not rely only on covariance distance; subspace mismatch appears more informative for deciding when an older state is worth retrieving. A

468 future gate should likely combine subspace features with
469 distance margin, previous-source rank, temporal gap,
470 session quality, decoder confidence, and perhaps hidden-
471 state or logit-space diagnostics.

472 The library-size ablation further suggests that mem-
473 ory need not be unbounded. Under the current
474 covariance-nearest rule, last-5 selection matches all-past
475 selection at $K = 20$, while previous-only remains the
476 best aggregate baseline. This points toward a compact
477 state library with a conservative retrieval policy: keep a
478 small set of recent or representative input states, default
479 to recency, and use geometry only when multiple signals
480 agree that an older state is likely to be better.

481 This framing also clarifies how the work relates
482 to stronger decoder-backbone approaches. Pretrained
483 speech models such as Wav2Vec2 may improve the repre-
484 sentational power of future brain-to-text decoders, while
485 our results address a different deployment problem: even
486 with a fixed decoder backbone, the input mapping can
487 drift across sessions and must be selected or adapted ef-
488 ficiently. Thus backbone improvements and input-state
489 adaptation are complementary.

490 The inner-speech feasibility result suggests the same
491 memory view may apply across behaviors. Future input-
492 state libraries may need to index both recording state
493 and intended speech mode, because a new neural geom-
494 etry may reflect session drift, speech-mode shift, or both.

495 7 Limitations

496 This study is a decoder-facing audit, not a new de-
497 coding architecture. Greedy PER is lighter than full
498 WER and avoids the language-model/Redis evaluation
499 stack. T12 is geometry-only and should not be inter-
500 preted as T12 PER replication. The inner-speech anal-
501 ysis is also geometry-only; it does not decode imagined
502 speech or evaluate a communication system. The learned
503 adapter and K-shot source-selection experiments simu-
504 late a beginning-of-day calibration window within the
505 validation split and evaluate only on the remaining tri-
506 als. They are intended to map adaptation rungs, not to
507 optimize a final deployment benchmark. We focus the
508 main claims on $K \leq 20$; larger calibration windows can
509 be informative as feasibility checks, but in this valida-
510 tion split they consume a large fraction of the available
511 session and are therefore not treated as minimal calibra-
512 tion.

513 8 Conclusion

514 For T15 brain-to-text decoding, fixed non-native in-
515 put layers break the decoder, moment correction is too
516 weak, and learned input adaptation recovers only part
517 of the lost performance. Most of the practical recov-
518 ery comes from selecting a compatible past input layer.
519 The immediately previous session is a strong default,
520 while drift geometry is a measurable signal for deciding
521 when that default may need to be overridden. This sup-
522 ports a recency-aware, geometry-guided adaptation lad-
523 der: start with the previous input state, measure drift

geometry from a short unlabeled window, and move to 524
stronger calibration only when the geometry indicates 525
that simple source selection is not enough. 526

Ethical Statement 527

This work uses publicly released intracortical BCI 528
datasets and reports only aggregate analyses. It does not 529
attempt to infer private inner speech or deploy a decoder. 530
Future extensions to inner-speech or cross-behavior state 531
selection should include explicit intent-gating and pri- 532
vacy safeguards. 533

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